

Problem 9

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Problem 1 (a) Given function f on an interval $[a, b]$:

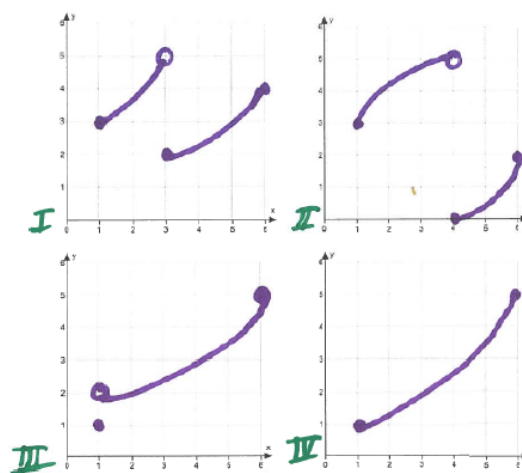
(i) What are the conditions of the Intermediate Value Theorem?

Explanation. f is continuous on the interval $[a, b]$

(ii) What is the conclusion of the Intermediate Value Theorem?

Explanation. For any number L strictly between $f(a)$ and $f(b)$, there is at least one number c in (a, b) satisfying $f(c) = L$. This means that for any L strictly between $f(a)$ and $f(b)$, the horizontal line $y = L$ intersects the graph of f .

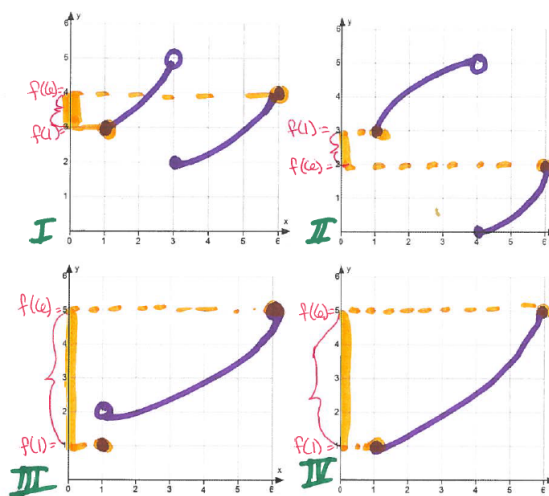
(b) Given the four functions on the interval $[1, 6]$, answer the questions below.



(i) For each of the functions I through IV, indicate $f(1)$ and $f(6)$. Then mark the interval of all numbers strictly between $f(1)$ and $f(6)$, on the y-axis.

Explanation.

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- (ii) For each of the functions I through IV, write an interval of all numbers strictly between $f(1)$ and $f(6)$

Explanation. $f = I$: $(3, 4)$

$f = II$: $(2, 3)$

$f = III$: $(1, 5)$

$f = IV$: $(1, 5)$

- (iii) List the functions that satisfy the conditions of the Intermediate Value Theorem on $[1, 6]$

Explanation. Only function IV is continuous on $[1, 6]$

- (iv) For which of the functions is the following statement true: For any number L strictly between $f(1)$ and $f(6)$, there exists a number c in $(1, 6)$ satisfying $f(c) = L$.

Explanation. The statement is true for functions I and IV

- (v) Does the function III satisfy the conclusion of the Intermediate Value Theorem? Why or why not?

Explanation. No, function III does not satisfy the conclusion of the IVT. For example, take $L = 1.5$. $f(c) = 1.5$ has no solution in $(1, 6)$. Draw the horizontal line $y = 1.5$. What do you notice? This line does not intersect the graph of function III. What does this mean? The function does not attain that value, 1.5. So there is no c in $(1, 6)$ such that $f(c) = 1.5$.